Difficulties of Unconfoundedness in Observational Studies for Causal Effects

Chapter 16, *A First Course in Causal Inference* (2023) by Peng Ding

Presented by Sweta Karlekar & Gabriel Chuang (Columbia CS)

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Overview

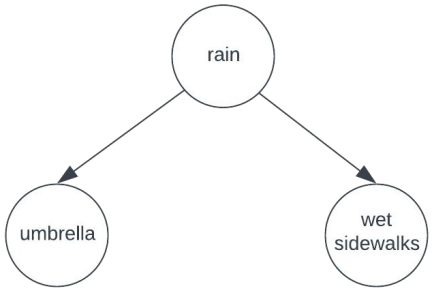
1. Basics of a causal diagram
    a. Dependency flow
    b. Confounding
2. Assessing unconfoundedness assumption
    a. Negative outcomes
    b. Negative exposures
3. Problems of over-adjustment
    a. M-bias
    b. Z-bias

Basics of a causal diagram

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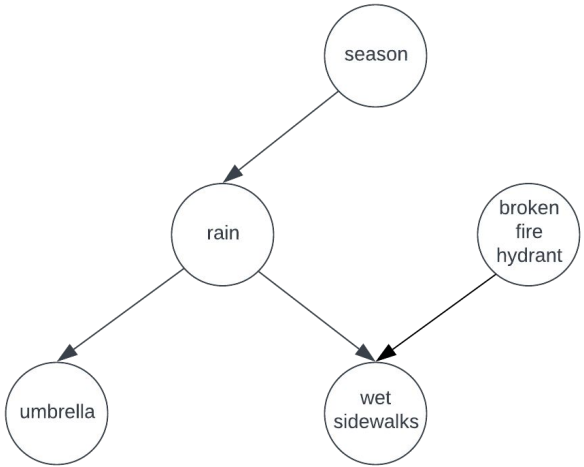
Basics of a causal diagram

- ➔ Flow of causality
  - ◆ Given by directed edges in a causal diagram (non-symmetric)
- ➔ Flow of dependence / flow of association
  - ◆ “If we already have some information about node A, can this give us a better estimate of node B?”
  - ◆ Symmetric
  - ◆ This path of dependence can be “blocked” given certain circumstances; when this happens, we say nodes are **d-separated**



Basics of a causal diagram

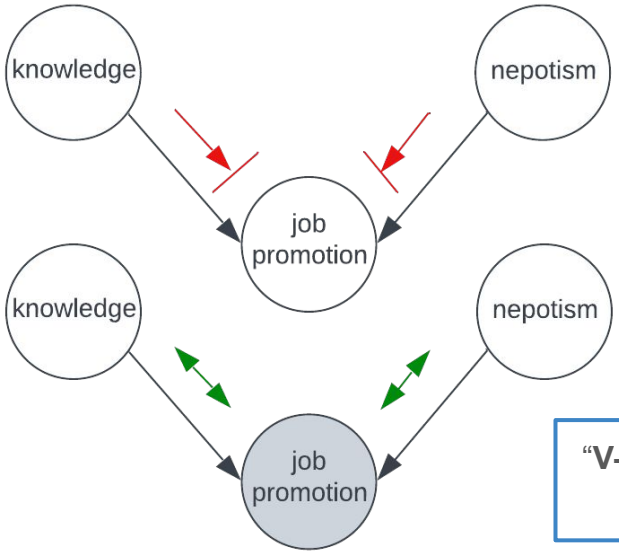
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Dependency flow

We can use this to quickly determine from a graphical model if two nodes are independent (d-separated) or not.

Case 1.



A path is <u>blocked</u> if and only if:

1. **It contains a collider that has not been conditioned on and has no descendants that have been conditioned on.**
2. It contains a non-collider that has been conditioned on

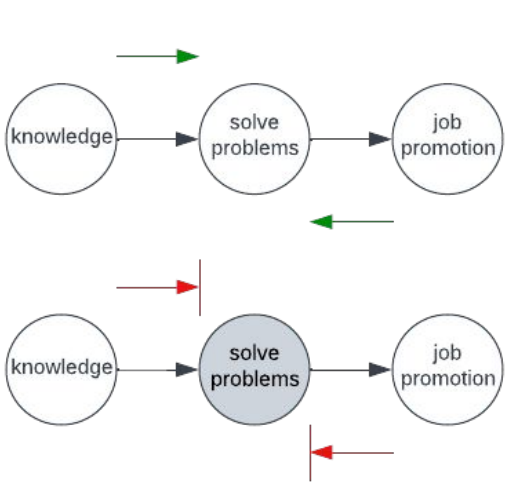
"**V-structure**" or "**collider**": Child node with unconnected parents

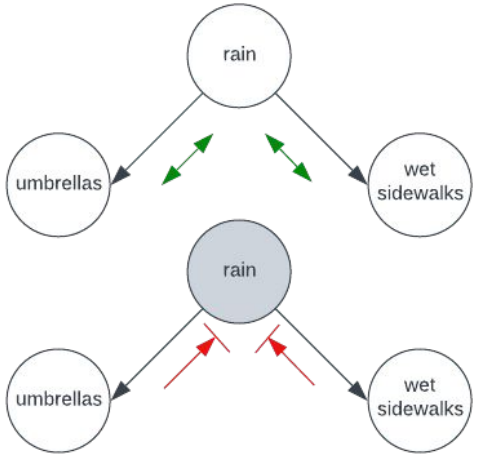
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Dependency flow

We can use this to quickly determine from a graphical model if two nodes are independent (d-separated) or not.

Case 2.



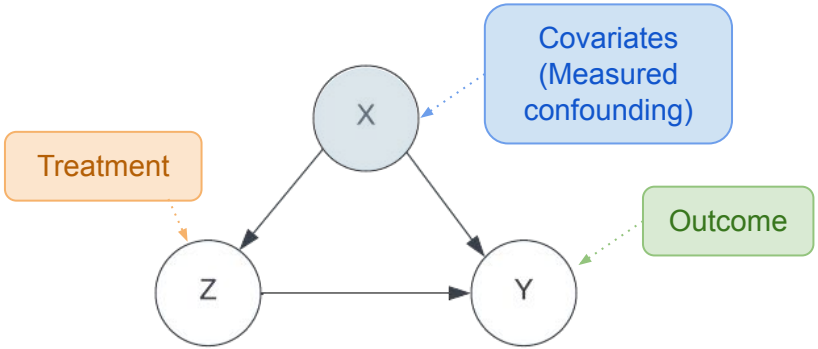


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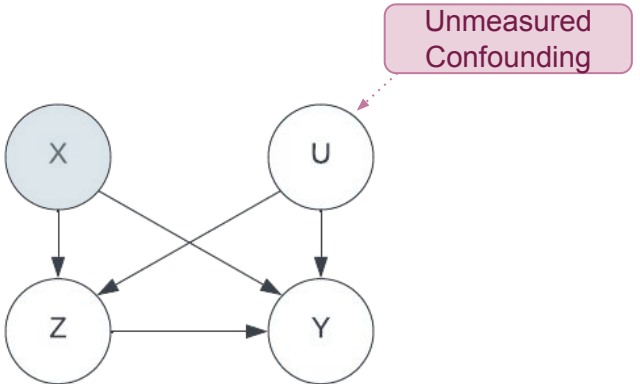
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Confounding



$$X \sim F_X(x)$$
$$Z = f_Z(X, \varepsilon_Z)$$
$$Y(z) = f_Y(X, z, \varepsilon_Y(z))$$

$$Z \perp\!\!\!\perp Y(z) \mid X$$



$$X \sim F_X(x)$$
$$U \sim F_U(u)$$
$$Z = f_Z(X, U, \varepsilon_Z)$$
$$Y(z) = f_Y(X, U, z, \varepsilon_Y(z))$$

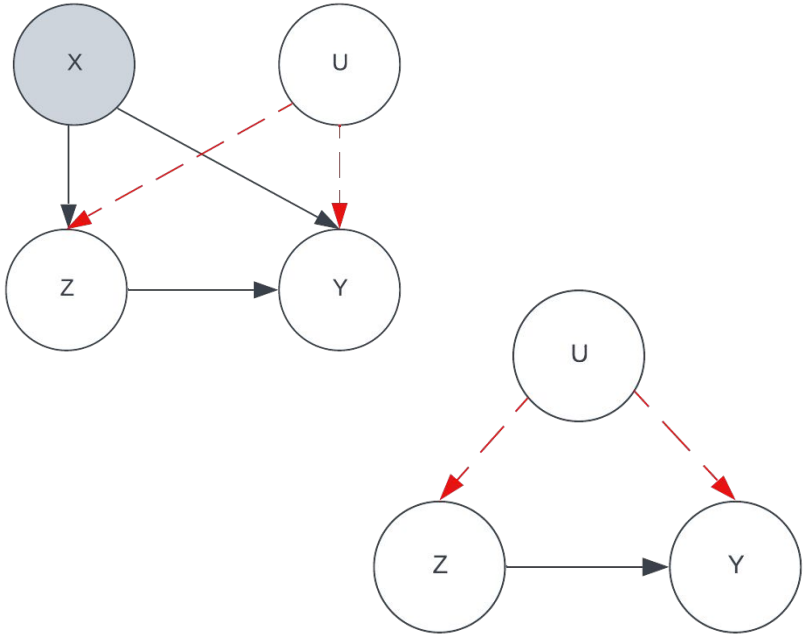
$$Z \perp\!\!\!\perp Y(z) \mid (X, U)$$
$$Z \not\perp\!\!\!\perp Y(z) \mid X$$

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Assessing unconfoundedness assumption

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Scenario

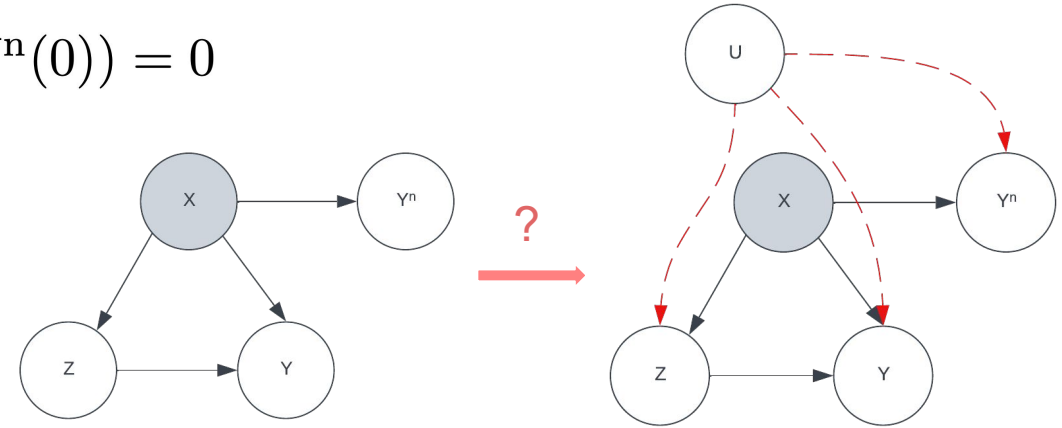


- ➔ Have you considered all the relevant confounders?
- ➔ **<u>There is no test for confounding</u>**, but there are ways to assess the validity of unconfoundedness
  - ◆ Negative outcomes
  - ◆ Negative exposures

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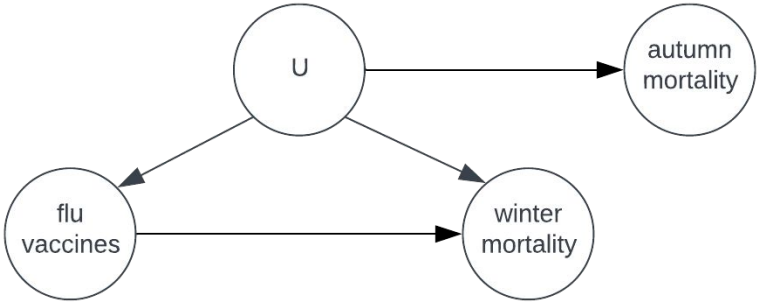
Negative outcomes

- Find a Y^n such that it is reasonable to assume that Y^n shares the same confounding structure as Y.
- $Z \perp\!\!\!\perp Y(z) \mid X \rightleftharpoons Z \perp\!\!\!\perp Y^n(z) \mid X$
- $\tau(Z \to Y^n) = \mathrm{E}(\mathrm{Y^n}(1) - \mathrm{Y^n}(0)) = 0$



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Negative outcomes (examples)

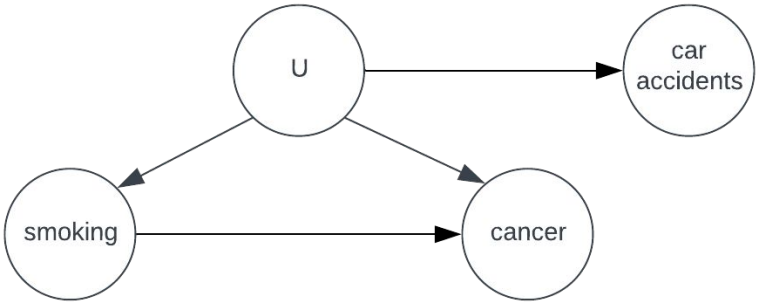


Jackson et al. (2006)

$$\tau(Z \to Y^n) = \mathrm{E}(\mathrm{Y^n}(1) - \mathrm{Y^n}(0)) > 0$$

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Negative outcomes (examples)

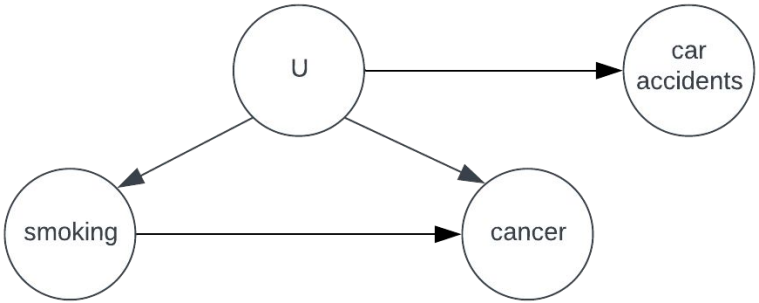


Cornfield et al. (1959)

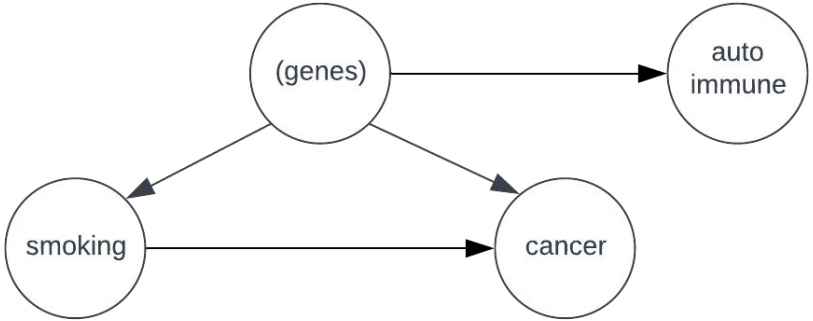
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Negative outcomes (examples)



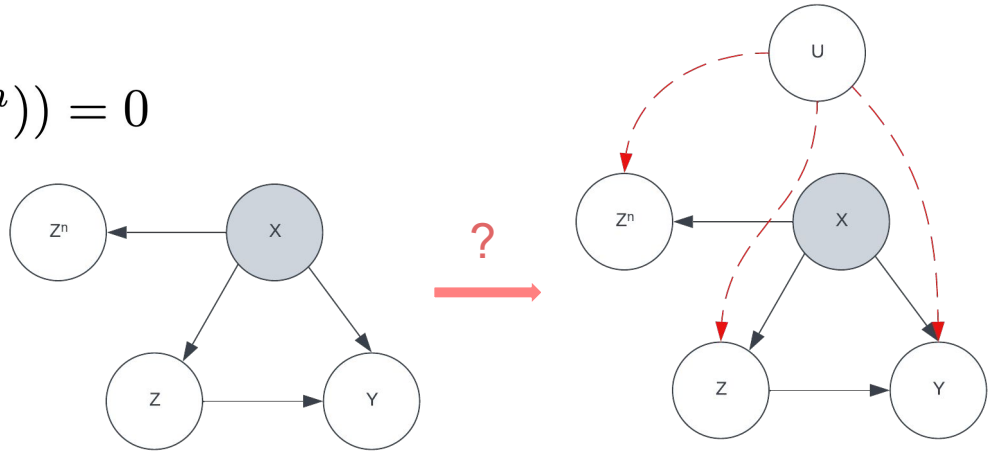
Cornfield et al. (1959)



(a potentially more modern and useful scenario)

Negative exposures

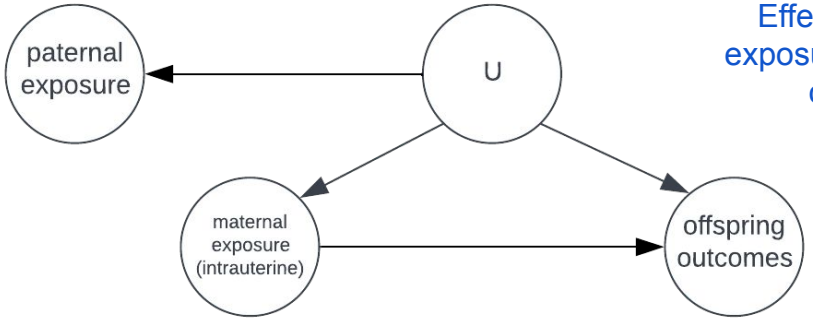
- Analog to negative outcomes
- Find a Z^n such that it is reasonable to assume that Z^n shares the same confounding structure as Z.
- $Z \perp\!\!\!\perp Y(z) \mid X \rightleftharpoons Z^n \perp\!\!\!\perp Y(z) \mid X$
- $\tau(Z^n \rightarrow Y) = \mathrm{E}(\mathrm{Y}(1^n) - \mathrm{Y}(0^n)) = 0$



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Negative exposures (examples)

$$\underbrace{\tau(Z^n \rightarrow Y) = \mathrm{E}(\mathrm{Y}(1^n) - \mathrm{Y}(0^n))}_{\text{Effect of paternal exposure on offspring outcomes}} < \underbrace{\tau(Z \rightarrow Y) = \mathrm{E}(\mathrm{Y}(1) - \mathrm{Y}(0))}_{\text{Effect of maternal exposure on offspring outcomes}}$$



Sanderson et al. (2017)

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Practical considerations

- ➔ Negative outcomes or negative exposures can only strengthen or weaken evidence for the unconfoundedness assumption; it cannot **test** it
- ➔ Researchers need to be very familiar with domain-specific potential causal problems in a study to find convincing negative exposures or outcomes

Problems of over-adjustment

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When should we condition on covariates?

"There is no reason to avoid adjustment for a variable describing subjects before treatment" ~ Rosenbaum (2002b)

"Typically, the more conditional an assumption, the more acceptable it is." ~ Rubin (2007)

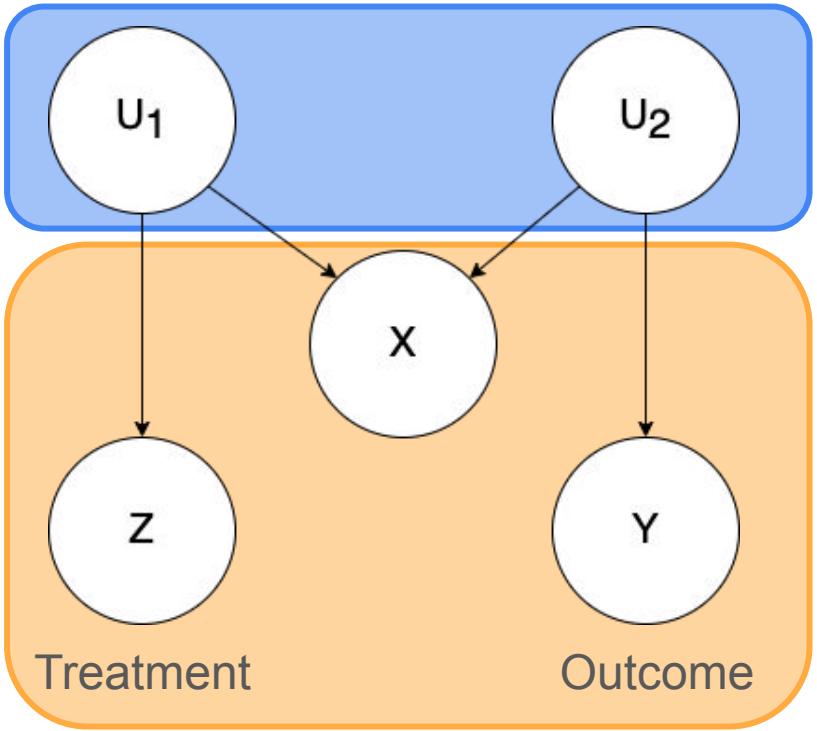
Should we always control for observed pre-treatment covariates when available?

A: No!

➔ M-bias
➔ Z-bias

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M-bias (Pearl 2009)



Unobserved

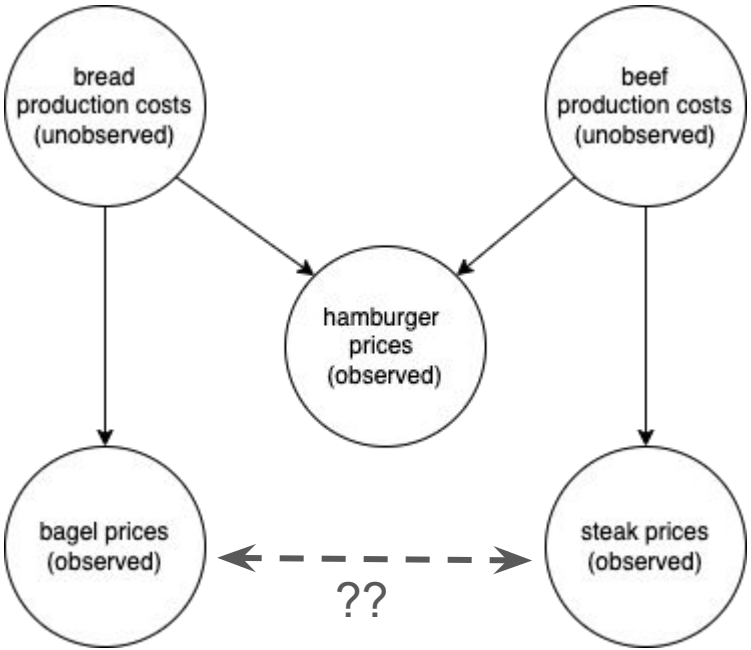
Observed

$$U_1 \perp\!\!\!\perp U_2,$$
$$X = f_X(U_1, U_2, \varepsilon_X),$$
$$Z = f_Z(U_1, \varepsilon_Z),$$
$$Y = Y(z) = f_Y(U_2, \varepsilon_Y)$$

Should we condition on X?

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M-bias (example)

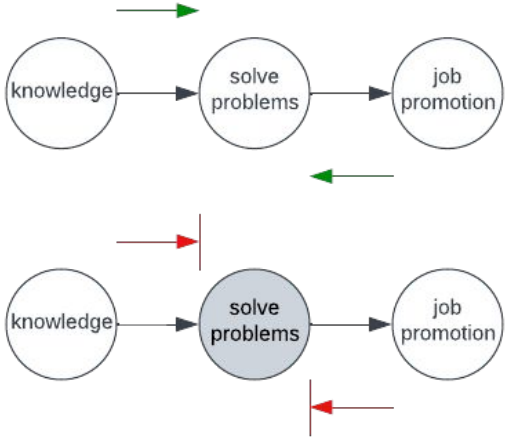


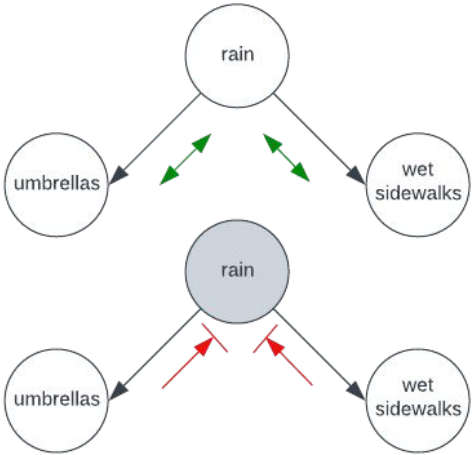
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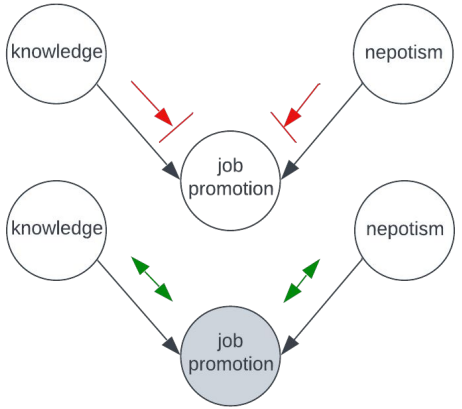
Recall – Dependency flow

A path is blocked if and only if:

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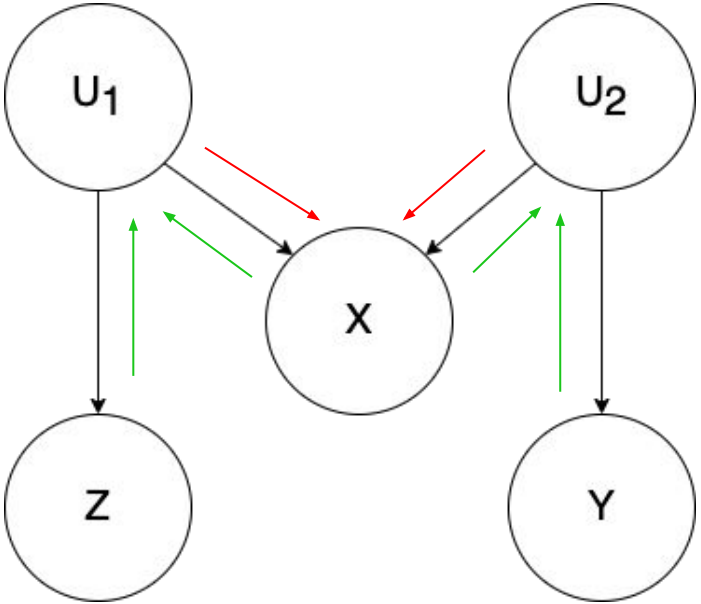


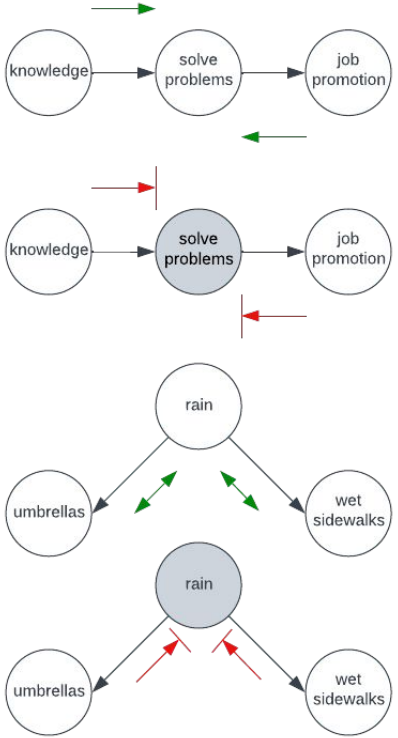


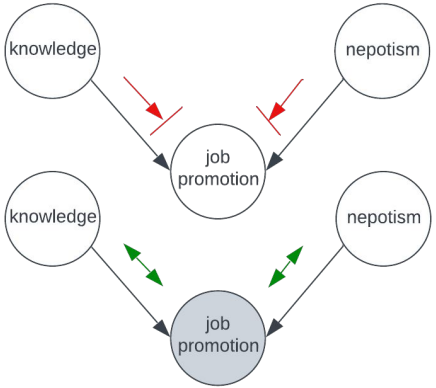


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Without conditioning, are Z and Y independent?



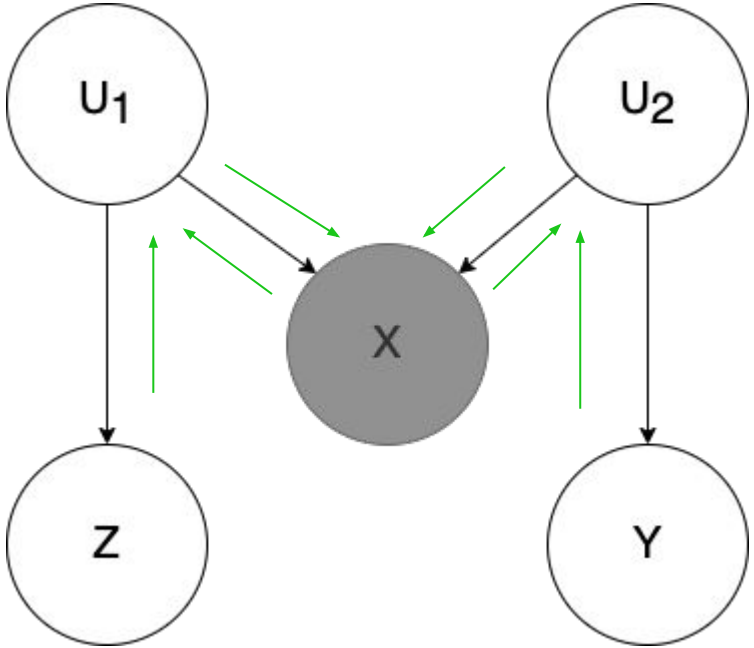


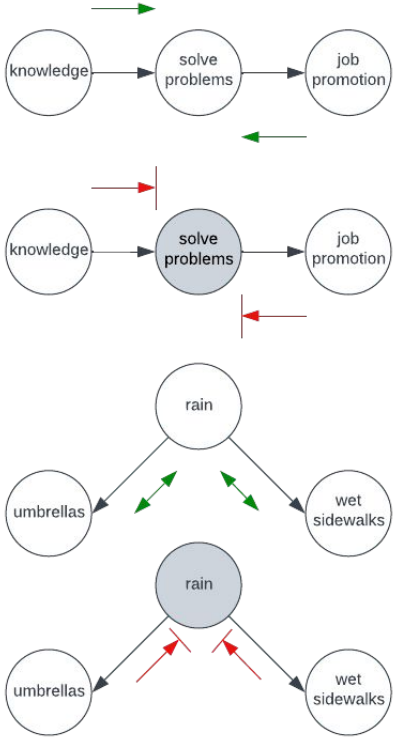


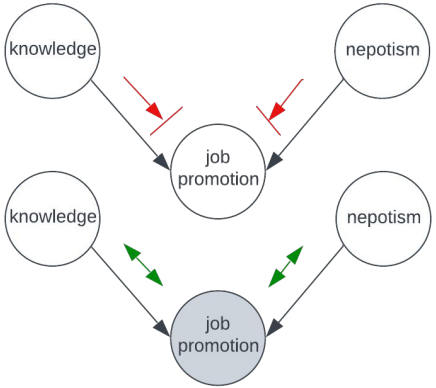
$$Z \perp\!\!\!\perp Y$$

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Conditioning on X, are Z and Y independent?



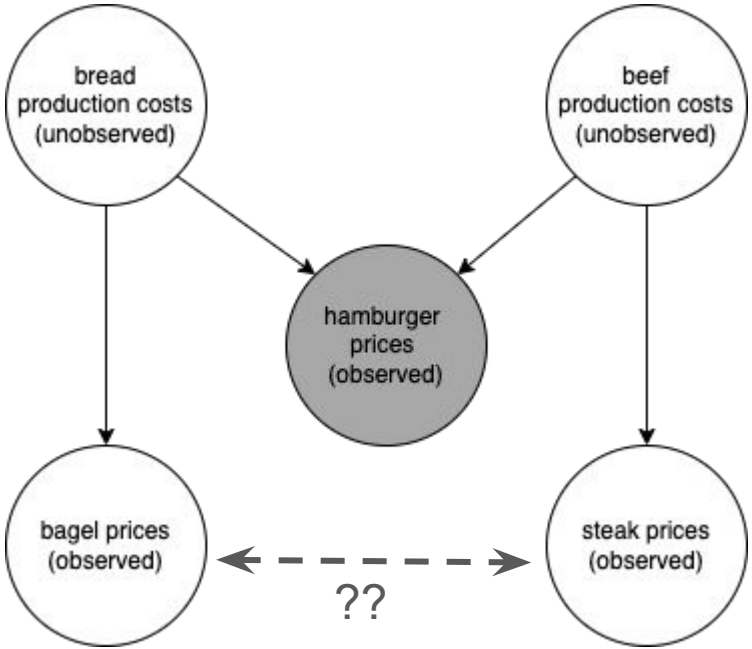




$$Z \not\perp Y \mid X$$

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M-bias (example)



condition on X (hamburger prices) → dependency flow through M shape

Fixing hamburger prices, what does an increase in bagel prices tell you about steak prices?

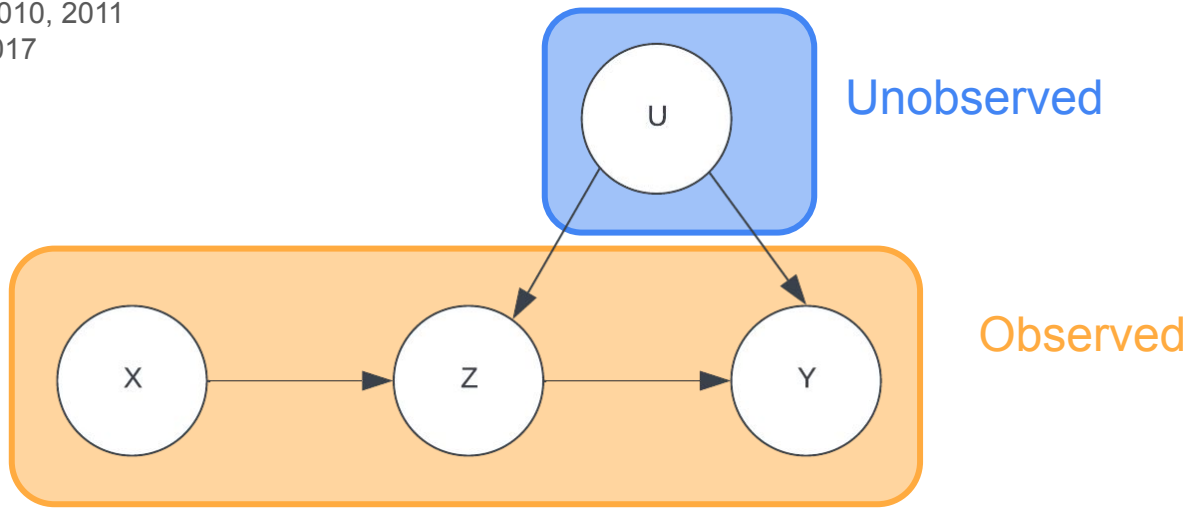
=> If the causal diagram has an M shape, you should *not* condition on X!

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Z-bias/Instrumental Variable bias

Observed by:

- Heckman/Navarro-Lozano 2004
- Wooldridge 2016
- Pearl 2010, 2011
- Ding 2017

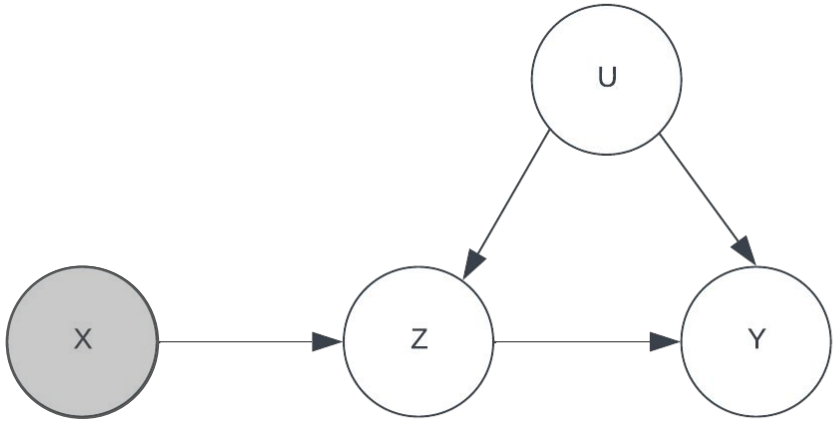


Should we condition on X?

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Z-bias intuition

$$X \sim N(0, 100)$$
$$U \sim N(0, 1)$$
$$Z = X + U + \varepsilon_Z$$
$$Y = Z + U + \varepsilon_Y$$



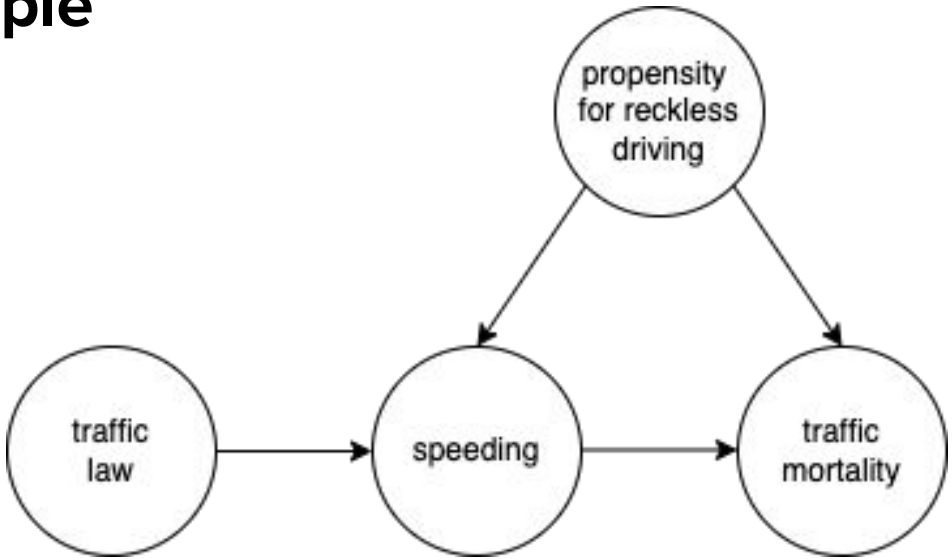
There is:

- strong causal effect along X → Z → Y
- weak confounding from U

What happens when you condition on X?

Y becomes a function of only U (and error terms)!

Z-bias example

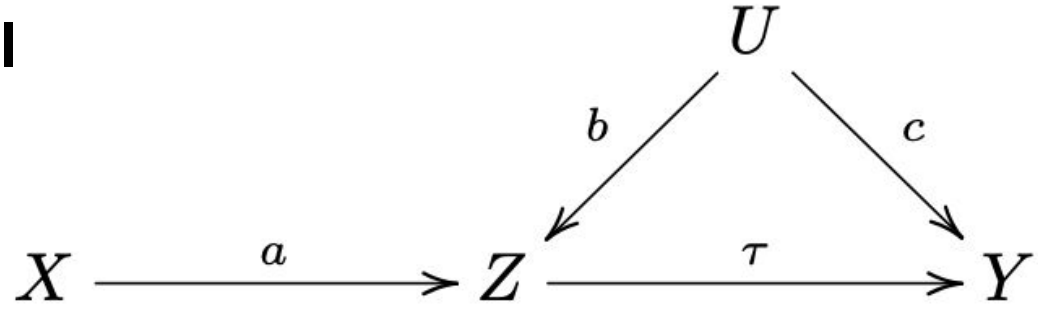


If we condition on traffic law, what do we observe?

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Z-bias in a linear model

$$\begin{cases} Z = aX + bU + \varepsilon_Z, \\ Y(z) = \tau z + cU + \varepsilon_Y, \end{cases}$$



(with X, U, and epsilons all independent standard normals)

Without conditioning on X:

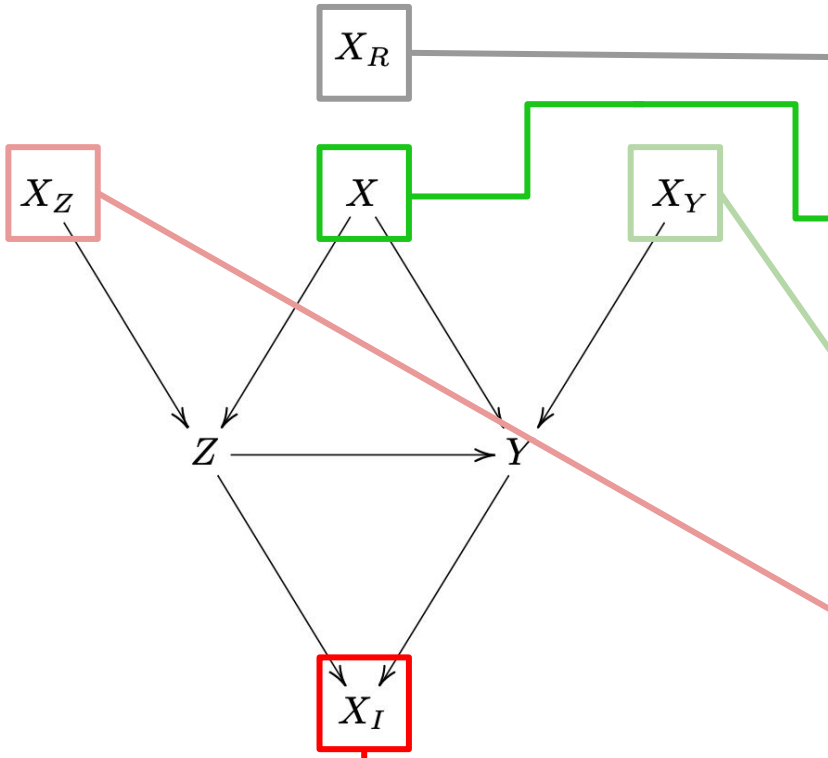
$$\begin{aligned} \tau_{\text{unadj}} &= \frac{\text{cov}(Z,Y)}{\text{var}(Z)} \\ &= \frac{\text{cov}(Z, \tau Z + cU)}{\text{var}(Z)} \\ &= \tau + \frac{c\text{cov}(aX + bU, U)}{\text{var}(Z)} \\ &= \tau + \frac{cb}{a^2 + b^2 + 1}, \end{aligned}$$

Conditioning on X:

$$\tau_{\text{adj}} = \tau + \frac{bc}{b^2 + 1}$$

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Which covariates should we condition on?



It doesn't matter (no effect)

Yes! X is a confounder; **must** condition on X to ensure unconfoundedness

Not necessary; but can help precision

Generally, no.
- if all confounders are observed, conditioning increases variance
- if any confounders are unobserved, conditioning **causes Z-bias**

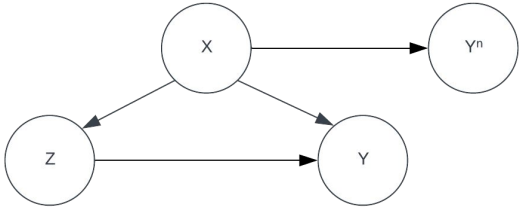
No; conditioning opens a path for dependency flow from Z to Y

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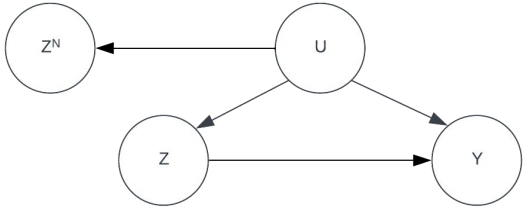
Recap

How can you assess how much the unconfoundedness assumption holds?

Negative outcomes

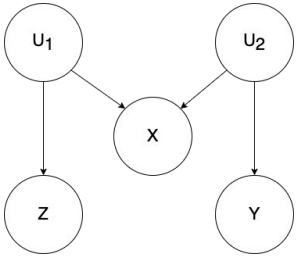


Negative exposures

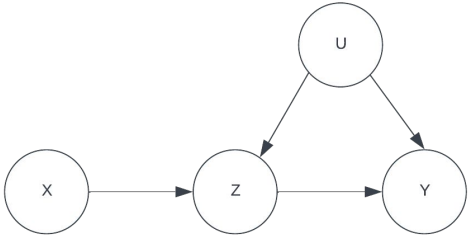


When is it bad to condition on observed covariates?

M-bias



Z-bias



It's important to have domain knowledge to have correct causal graphs!

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Questions?

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